# Unique repertoire of anti-carbohydrate antibodies in individual human serum

Search within citing articles

## Research progress in RBC alloimmunization

Z Fu, X Hao, W Gao, Q Wang - Frontiers in Immunology, 2025 - frontiersin.org

Red blood cell (RBC) alloimmunization is a common and clinically significant immunological phenomenon in transfusion medicine, pregnancy management, and organ transplantation. It …

☆ Save 99 Cite Cited by 14 Related articles All 6 versions »

## Insights into glycobiology and the protein-glycan interactome using glycan microarray technologies

J Heimburg-Molinaro, AY Mehta, CA Tilton… - Molecular & Cellular …, 2024 - mcponline.org

Glycans linked to proteins and lipids and also occurring in free forms have many functions, and these are partly elicited through specific interactions with glycan-binding proteins …

☆ Save 99 Cite Cited by 33 Related articles All 6 versions

## Mammalian cell-based production of glycans, glycopeptides and glycomodules

T Jaroentomeechai, R Karlsson, F Goerdeler… - Nature …, 2024 - nature.com

Access to defined glycans and glycoconjugates is pivotal for discovery, dissection, and harnessing of a range of biological functions orchestrated by cellular glycosylation …

☆ Save 99 Cite Cited by 19 Related articles All 13 versions »

## Serum antibody screening using glycan arrays

S Marglous, CE Brown, V Padler-Karavani… - Chemical Society …, 2024 - pubs.rsc.org

Humans and other animals produce a diverse collection of antibodies, many of which bind to carbohydrate chains, referred to as glycans. These anti-glycan antibodies are a critical part …

☆ Save 99 Cite Cited by 32 Related articles All 6 versions

## Human anti-glycan reactivity emerges from B cells utilizing private gene rearrangements that are affinity maturated in germinal centers

JS New, CF Fucile, AR Callahan, JN Burke, RS Davis… - Immunity, 2026 - cell.com

The human antibody repertoire includes reactivity toward a broad array of carbohydrate antigens associated with commensal microflora and pathogenic organisms. Here, we …

☆ Save 99 Cite Cited by 5 Related articles All 5 versions

## ABO blood groups and galectins: Implications in transfusion medicine and innate immunity

CM Arthur, M Hollenhorst, SC Wu, R Jajosky… - Seminars in …, 2024 - Elsevier

ABO blood group antigens, which are complex carbohydrate moieties, and the first human polymorphisms identified, are critical in transfusion medicine and transplantation. Despite …

☆ Save 99 Cite Cited by 22 Related articles All 7 versions

## *O*-glycosylation of IgA1 and the pathogenesis of an autoimmune disease IgA nephropathy

J Novak, RG King, J Yother, MB Renfrow… - …, 2024 - academic.oup.com

IgA nephropathy is a kidney disease characterized by deposition of immune complexes containing abnormally O-glycosylated IgA1 in the glomeruli. Specifically, some O-glycans …

☆ Save 99 Cite Cited by 20 Related articles All 9 versions

## Novel click coupling chemistry to explore glycan recognition

T Jia, AY Mehta, CA Tilton, EKC Tulin… - ACS Central …, 2025 - pmc.ncbi.nlm.nih.gov

Specific recognition of glycans by proteins is important in many biological processes and immune responses. Here we present a general approach for derivatizing free glycans with a …

☆ Save 99 Cite Cited by 9 Related articles All 7 versions

## Offsetting low-affinity carbohydrate binding with covalency to engage sugar-specific proteins for tumor-immune proximity induction

BPM Lake, AF Rullo - ACS Central Science, 2023 - pmc.ncbi.nlm.nih.gov

Carbohydrate-binding receptors are often used by the innate immune system to potentiate inflammation, target endocytosis/destruction, and adaptive immunity (eg, CD206, DC-SIGN …

☆ Save 99 Cite Cited by 13 Related articles All 6 versions

## Antibody diversity in IVIG: Therapeutic opportunities for novel immunotherapeutic drugs

S von Gunten, C Schneider, L Imamovic… - Frontiers in …, 2023 - frontiersin.org

Significant progress has been made in the elucidation of human antibody repertoires. Furthermore, non-canonical functions of antibodies have been identified that reach beyond …

☆ Save 99 Cite Cited by 18 Related articles All 9 versions »